

**“ACADEMIC RULES AND REGULATIONS OF THE NSUT  
B.TECH. PROGRAMME”**

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**1. CURRICULUM STRUCTURE**

**1.1.** B.Tech. programme of the University shall be based upon CBCS and shall have well defined Programme Educational Objectives (PEOs).

**1.2.** All the courses shall have well-defined Course Outcomes (COs).

**1.3.** B.Tech. programmes shall have a *Semester-wise Course* scheme with detailed syllabi and an evaluation scheme of the various courses duly approved by the BoM on the recommendations of the BoS of the various Departments and the Senate.

**1.4.** A course may be designed to comprise lectures, tutorials, laboratory work, field work, outreach activities, project work, vocational training,

viva, seminars, term papers, assignments, presentations etc. or a combination of some of these components.

**1.5.** Courses shall be of three kinds: Core, Elective and Foundation.

a. **Core Course (CC):** This is a course which is to be compulsorily studied by a student as a core requirement to complete the requirement of the B.Tech. programme.

b. **Elective Course:** This is a course which can be chosen from a pool of elective courses. It is intended to support the discipline of study by providing an expanded scope, enabling exposure to another discipline/domain and nurturing a student's proficiency and skill. An elective may be of the following types:

i. **Discipline Centric Elective (ED):** It is an elective course that adds proficiency to the students in the discipline.

ii. **Generic Elective (EG):** It is an elective course taken from other engineering subjects and enhances the generic proficiency and interdisciplinary perspective of students.

iii. **Open Elective (EO):** It is an elective course taken from a common pool of non-engineering disciplines that broadens the perspective of an engineering student. These electives shall comprise two groups: Open electives of the Humanities, Social Sciences and Management group and Open electives of the Sciences group.

c. **Foundation Course:** A Foundation course leads to knowledge enhancement and provides value-based training. Foundation courses may be of two kinds:

i. **Compulsory Foundation (FC):** It is based upon the content that leads to fundamental knowledge enhancement in Sciences, Humanities, Social Sciences and Basic engineering. They are mandatory for all disciplines.

ii. **Elective Foundation (FE):** It can be taken from among a common pool of foundation courses which aim at value-based education. They may provide hands-on training to improve competencies,

skills or provide education on human, societal, environmental and national values. These shall be mandatory, non-credit courses, which do not carry any credits but a student has to pass in order to be eligible for award of degree.

**1.6.** The requirements for the various types of courses i.e. FC/CC/FE/ED/EG and EO shall be given in the *Semester-wise Course* scheme for all B.Tech. programmes.

**1.7.** A course may have pre-requisite course(s) which shall be specified in the *Semester-wise Course scheme*.

**1.8.** Each course contributes certain credits to the programme. A course can be offered either as a full course (4 credits) or as a half course (2 credits). A full course is conducted with 3 hours of lectures and either 1 hour of tutorial or 2 hours of practical work per week. A half course is conducted with 2 hours of lectures or 4 hours of practical work.

**1.9.** A student of the programme has to accumulate at least 50% credits from Core courses; about 20% credits from Foundation courses; and the remaining credits from Elective courses, to become eligible for award of the degree.

**1.10.** During the span of the programme the student shall earn about 12 credits of Humanities, Social Sciences and Management, 24 credits of Sciences, 24 credits of other engineering disciplines and 16 credits from other elective courses. The remaining credits shall be earned from the discipline-specific core/elective courses.

**1.11.** A project is considered as a special course involving application of the knowledge gained during the course of study in exploring, analysing and solving complex problems. A candidate shall complete such a course with advisory support by one or more University faculty members. It is mandatory to pass the project courses to become eligible for award of the degree.

**1.12.** Apart from the above courses, Audit courses may be offered in summer vacations. These courses do not carry credits but they aim at expanding the knowledge or bridging deficiency in knowledge or skills.

## **2. PROGRAMME DURATION AND STRUCTURE**

**2.1.** An academic year shall be apportioned in two semesters, namely odd and even. Each semester shall consist of approximately eighteen weeks.

**2.2.** The duration of the programme shall not be less than eight regular semesters (four years) and the maximum span of the course shall be seven years.

**2.3.** There shall be an academic calendar for each semester. The schedule of academic activities including the dates of registration, MSE/ESE shall be governed by the Academic Calendar. The calendar shall also specify dates during which the co-curricular and extra-curricular activities may be organized.

**2.4.** A student has to register for the requisite number of courses before the start of a semester as per the schedule given in the Academic Calendar and guidelines issued by the office of the Dean, Academics from time to time.

**2.5.** The University may cancel the registration of all the courses in a given semester if-

- a. The student has not cleared the dues to the University/hostel.
- b. A punishment is awarded leading to cancellation of the student's registration.

**2.6.** A student can opt for a course only if he/she has successfully passed its pre-requisite(s), wherever applicable/specified.

**2.7.** A student may register for courses leading to a minimum number of credits as prescribed in the Semester-Wise Course scheme subject to the maximum credits specified therein (including backlog courses , if any).

**2.8.** The programme would consist of 170 credits. A student shall be awarded the degree if he/she has earned 162 or more credits. The CGPA shall be calculated at the end of the programme, on the basis of the best 162 credits. These credits shall not be earned through on-line courses.

**2.9.** A student shall be awarded the degree with honours if he/she has earned 182 credits with at least 8.50 CGPA. The additional 12 credits may be earned through University-recommended on-line courses only (for example the equivalent credits offered by NPTEL of 4 week, 8 week and 12 week on-line courses shall be of 2, 3 and 4 credits respectively). The registration fees shall be borne by the student. These on-line courses shall be cleared within first four years (not necessary one course in each semester). After successful completion of these on-line courses the students shall provide their successful completion status/certificates to the CoE through their respective Departments.

### **3. EVALUATION AND ASSESSMENT**

The performance of a student in a semester shall be evaluated through continuous class assessment, MSE and ESE. Both the MSE and ESE shall be University examinations and will be conducted as notified by the CoE of the University. The marks for continuous assessment (Sessional marks) shall be awarded at the end of the semester. The continuous assessment shall be based on class tests, assignments/tutorials, quizzes/viva-voce and attendance etc. The MSE/ESE shall comprise of written papers, practicals and viva-voce, inspection of certified course work in classes and laboratories, project work, design reports or by means of any combination of these methods.

The weightage of each of these modes of evaluation for the different types of courses shall be as per Table 1. Further, the mechanism for continuous assessment shall be as per Table 2.

**Table-1: Evaluation Scheme**

| S. No.                                                                                                                                | Type of Course                       | Continuous Assessment (CA) Theory      | Mid-Semester Examination (MSE) Theory | End-Semester Examination (ESE) Theory | Continuous Assessment (CA) Practical | End-Semester Examination (ES) Practical |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------|---------------------------------------|---------------------------------------|--------------------------------------|-----------------------------------------|
| 1                                                                                                                                     | FE courses                           | Continuous Assessment only (100 marks) |                                       |                                       |                                      |                                         |
| 2                                                                                                                                     | CC/FC/ED/EG/EO Theory with Tutorial  | 25                                     | 25                                    | 50                                    | Nil                                  | Nil                                     |
| 3                                                                                                                                     | CC/FC/ED/EG/EO Theory with Practical | 15                                     | 15                                    | 40                                    | 15                                   | 15                                      |
| 4                                                                                                                                     | Project I and Project II             | Nil                                    | Nil                                   | Nil                                   | 40                                   | 60                                      |
| 5                                                                                                                                     | Training                             | Nil                                    | Nil                                   | Nil                                   | 40                                   | 60                                      |
| 6                                                                                                                                     | Audit Courses*                       | -                                      | -                                     | -                                     | -                                    | -                                       |
| *The distribution of marks of practical and/or theory components for Audit courses shall be determined by the respective Departments. |                                      |                                        |                                       |                                       |                                      |                                         |

**Table 2: Continuous Assessment**

| S. No. | Type of Course                       | Continuous Assessment (CA)                                                          |
|--------|--------------------------------------|-------------------------------------------------------------------------------------|
| 1      | CC/FC/ED/EG/EO Theory with Tutorial  | Two class tests, Assignments, Teachers' assessment (quizzes, viva-voce, attendance) |
|        | CC/FC/ED/EG/EO Theory with Practical | One class test, One Lab test, Assignments/Projects, Teachers' assessment            |
| 2      | FE courses                           | Two class tests, Assignments, Teachers' assessment                                  |
| 3      | Project I /II                        | Mid-Semester Presentation, Report, Supervisor's Assessment                          |
| 4      | Training                             | As specified by the Department                                                      |
| 5      | Audit Courses                        | As specified by the Department                                                      |

#### 4. GRADING

The relative grading system shall be implemented in awarding the grades and SGPA/CGPA under CBCS. A 10-point grading system and corresponding grade points shall be used with the letter grades as given in Table 3.

**Table 3: The Grades and the Grade Points**

| S. No. | Letter Grade | Letter Grade          | Grade point |
|--------|--------------|-----------------------|-------------|
| 1      | O            | Outstanding           | 10          |
| 2      | A+           | Excellent             | 9           |
| 3      | A            | Very Good             | 8           |
| 4      | B+           | Good                  | 7           |
| 5      | B            | Above average         | 6           |
| 6      | C            | Average               | 5           |
| 7      | D            | Pass                  | 4           |
| 8      | F            | Fail                  | 0           |
| 9      | FD           | Fail due to detention | 0           |
| 10     | Ab           | Absent                | 0           |
| 11     | W            | Withdrawal            | NIL         |

**4.1** The award of the grades shall be based on the marks out of 100, as per the distribution of the various components given in Table 1.

**4.2** For the class strength of more than 30, a relative grading system shall be implemented and the grades shall be allotted on the basis of normalized score as described below and Table 4.

$$NS \text{ (Normalise Score)} = \frac{\text{Marks Obtained} - \text{Average Marks obtained by the class}}{\text{Standard Deviation of the Marks obtained by the class}}$$

**Table 4: Grade Allocation based on the Normalized Score**

| S. No. | Lower Range of | Grade | Upper Range of |
|--------|----------------|-------|----------------|
|--------|----------------|-------|----------------|

|   | <b>NS</b> |                | <b>NS</b> |
|---|-----------|----------------|-----------|
| 1 | >1.5      | O              | --        |
| 2 | >1.0      | A <sup>+</sup> | ≤ 1.5     |
| 3 | >0.5      | A              | ≤ 1.0     |
| 4 | >0.0      | B <sup>+</sup> | ≤ 0.5     |
| 5 | >-0.5     | B              | ≤ 0.0     |
| 6 | >-1.0     | C              | ≤ -0.5    |
| 7 | >-1.5     | P              | ≤ -1.0    |
| 8 | --        | F              | ≤ -1.5    |

**4.3** The normalized score shall be rounded off to two decimal places.

**4.4** Grade Moderation shall be undertaken for the adjustment of the grade boundaries as indicated in Table 4. These grade boundaries may be adjusted as follows.

- No student can be awarded 'D' or better grade without securing at least **30%** aggregate marks in a course.
- No student can be awarded 'O' grade without securing at least **85%** aggregate marks in a course.
- No student can be awarded less than 'D' if he/she scores more than **40%** aggregate marks in a course.
- No student can be awarded less than 'O' if he/she scores more than **95%** aggregate marks in a course.
- In case of any difficulty, the absolute grading method suggested in Table 5 shall be used.
- The final grade boundaries shall be retained in records.

**4.5** For the class strength of less than 30, the grades shall be allotted on the basis of absolute grading as given in Table 5.

**Table 5: Grades for Absolute Marks.**

| <b>S. No.</b> | <b>Marks</b> | <b>Grade</b>   | <b>Marks</b> |
|---------------|--------------|----------------|--------------|
| 1             | ≥ 90         | O              | ≤ 100        |
| 2             | ≥ 81         | A <sup>+</sup> | < 90         |
| 3             | ≥ 72         | A              | < 81         |

|   |           |                |      |
|---|-----------|----------------|------|
| 4 | $\geq 63$ | B <sup>+</sup> | < 72 |
| 5 | $\geq 54$ | B              | < 63 |
| 6 | $\geq 45$ | C              | < 54 |
| 7 | $\geq 35$ | D              | < 45 |
| 8 |           | F              | < 35 |

#### 4.6 Interpretation of the Grades

- Fail grade:** A student obtaining Grade F/FD/Ab shall be considered as failed and shall be required to register for the course again. In case of the elective courses, if the student does not want to register again in an **elective course in which he/she has failed** (for EG, ED, EO, FE courses *but not for* CC or FC courses), then he/she can register afresh for a new elective course. However, grades F/FD, as applicable, shall be mentioned in the Grade Card even after passing the course.
- ‘FD’ grade:** The FD grade indicates fail due to shortage of attendance in a course.
- ‘W’ grade:** This refers to the withdrawal from the courses other than core courses. Withdrawal shall be allowed as per the guidelines issued from time to time by the Dean Academics.
- Audit/FE courses:** For audit courses, grades shall be indicated however this will not be counted for the computation of the SGPA/CGPA.

#### 4.7 Computation of the SGPA and CGPA.

- The SGPA is the ratio of the sum of the product of the number of credits and the grade points scored in all the courses taken in a semester (including back-log courses), to the sum of the number of credits of all the courses taken by a student, that is:

$$SPGA(S_j) = \frac{\sum (C_i \times G_i)}{\sum C_i}$$

where,  $S_j$  is the  $j^{th}$  semester,  $C_i$  is the number of credits of the  $i^{th}$  course of that semester and  $G_i$  is the grade point scored by the student in the  $i^{th}$  course.

- b. The CGPA is also calculated in the same manner taking into account the best 162 credit courses of the student taken over all the semesters of a programme, that is:

$$CPGA = \frac{\sum(C_i \times G_i)}{\sum C_i}$$

where,  $C_i$  is the number of credits of the  $i^{th}$  course and  $G_i$  is the grade point scored by the student in the  $i^{th}$  course.

- c. The SGPA and CGPA shall be rounded off to 2 decimal points.
- d. CGPA shall be converted into percentage of marks by multiplying it with 10. Both CGPA and percentage shall be mentioned on the final transcript.

## **5. ATTENDANCE AND DETENTION**

**5.1.** Students of the programme are expected to attend every lecture, tutorial and practical class scheduled for them.

**5.2.** The students must have a minimum attendance of 75% of the total number of classes including lectures, tutorials and practicals, held in a subject till MSE/ESE in order to be eligible to appear in the MSE/ESE for that subject.

**5.3.** The Dean Academics, may allow relaxation in the minimum requirement of attendance upto 10% for reasons to be recorded. This relaxation may be granted on the production of documents showing that the student was either busy in any authorized activities or was absent due to medical/other genuine reasons. The student should submit these documents to the HoD, within seven days of resuming the studies. Certificates submitted later will not be considered.

**5.4.** Under exceptional circumstances, the Dean Academics may further relax the minimum attendance up to 5% on recommendation of a committee comprising of Dean Student Welfare, Dean of Faculty and HoD of the respective department.

**5.5.** Relaxation in attendance maybe granted for a maximum of 2 times during the duration of the programme.

**5.6.** A student shall not be permitted to appear in the MSE/ESE if his/her attendance till MSE/ESE is below 60 % after relaxation given in clauses 11.3 and 11.4.

**5.7.** Students who are not allowed to appear in the ESE due to shortage of attendance shall be awarded 'FD' grade. Such students shall have to register again for that course in subsequent years/summer semester to pass the course.

**5.8.** A student can register again for a different elective course in subsequent years/summer semester and pass the elective course.

**5.9.** The attendance shall be counted from the date of start of academic session. For first year students, attendance shall be counted from the date of the start of academic session or the actual date of admission, whichever is later.

## **6. PROMOTION AND PASSING A COURSE**

**6.1.** There shall not be any restriction on promotion from an odd semester to the next even semester.

**6.2.** For promotion from even semester to the next odd semester (i.e. of the next academic year) the student has to fully clear either of the semester of the academic year or earn credits greater than or equal to minimum credit of either of the semester of the academic year.

**6.3.** There shall be no supplementary examinations. A student who has failed in a course shall have to register again for the course in a subsequent year/summer semester.

**6.4.** If the student does not want to register again in an elective course

(that is, EG, ED, EO, FE but not CC or FC courses) then he/she can register again for a new elective course.

**6.5.** Summer semester may be run for back log courses in which there are 10 or more registrations. Separate grade card shall be issued for the summer semester (if applicable).

**6.6.** If a student wants to improve his/her grade in a course, he/she has to register again for the course. However, the student may register for a different elective course as per clause 12.4. Registration for improvement of grade in a course shall be allowed only once. However, the best grade in that particular course, shall be considered for computation of SGPA/CGPA.

**6.7.** To pass a course, the student should score at least 30% marks separately in the ESE of theory and practical (wherever applicable) components of the course.